Supplemental Material for

Design of frictionless interfaces for moiré layers

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I. Molecular dynamics simulation details

For the graphene on graphene interface in Fig.1(a), the moiré unit cell is defined by a twist vector $(m, n) = (22,23)$, and the relative twist angle between two layers is calculated as [1] $\theta = \arccos\frac{n^2+4mn+m^2}{2(n^2+mn+m^2)} = 1.47°$. The moiré periodicity is calculated as [2] $\lambda = \frac{|\overrightarrow{R_b}|}{2\sin\frac{\theta}{2}} = 95$ Å. For the graphene on h-BN heterostructure in Fig.1(d), with $p = 0.98$, the relative twist angle is $0.6°$, and moiré periodicity is $\lambda = \frac{p|\overrightarrow{R_b}|}{\sqrt{1+p^2-2p\cos\theta}} = 118$ Å.

Molecular dynamics simulations are performed using LAMMPS [3–9] to calculate friction forces across different sliding directions. For graphene on graphene homobilayers, intralayer carbon interactions are described by REBO potential [10], while interlayer vdW interactions are modeled using the Kolmogorov-Crespi (KC) potential [11]. For graphene on h-BN heterostructure, interlayer interactions are described by registry-dependent interlayer potential (ILP) with refined parameterization [12].

A timestep of 1fs is used. The temperature of the flake is controlled by a Nosé-Hoover thermostat under an NVT canonical ensemble at 0.1K. Fig. S1 demonstrates the atomic model of the graphene on graphene system ($\xi = 2/3$) visualized using OVITO [13]. A rigid virtual stage, duplicated from the top flake, initially shares the same spatial coordinates with the top flake. That is, each atom from the virtual stage coincides with a unique atom from the top flake. Each atom of the stage is then connected with its coinciding atom in sliding flake via a harmonic spring (with spring stiffness $K = 5eV/\text{Å}^2$). Following a 2 ps relaxation period, the sliding flake is dragged along a straight path in a specific direction by the stage, which is moved at a constant velocity of $10\ m/s$. This small velocity ensures full control of the flake's translation by the stage, preventing unintended motion in other directions without rigidifying the sliding flake.

From force equilibrium, $F_{total} = F_{spring} - f$. Therefore, the frictional force f can be calculated by subtracting the total force F_{total} felt by the flake from the spring force F_{spring} (drag force). Both F_{total} and F_{spring} are recorded throughout the sliding path. Since f fluctuates during sliding, we mainly focus on $F_{max} = \max(f)$ for each sliding direction to evaluate its frictional properties, as demonstrated in Figs. 1(b), 1(e).

For the calculations in Fig. 2(d), $\lambda = 95$ Å is used for graphene on graphene interface and $\lambda = 118$ Å for graphene on h-BN interface. For Fig. 2(e), $\xi_1 = 2/3$ and $\xi_2 = 1$. They both show results in the sliding direction of $\gamma = 90°$.

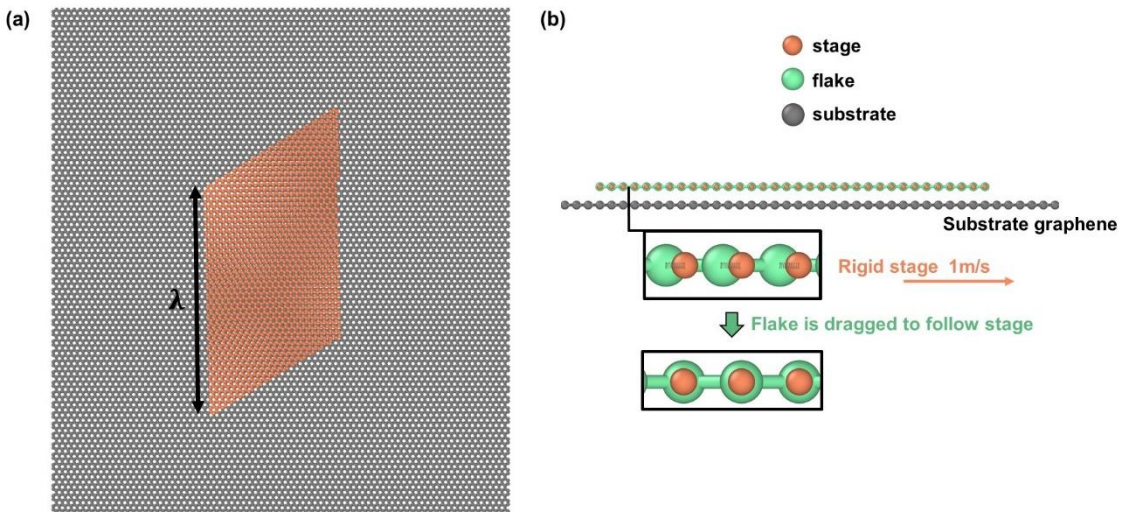

FIG. S1. Molecular dynamics (MD) simulation model of graphene on graphene interface. Orange, green and gray denote rigid virtual stage, sliding flake, and graphene substrate, respectively. (a) Top view and (b) side view. Atoms of the virtual stage and the flake are initially coincident and connected by harmonic springs with a stiffness of $K = 5eV/\text{Å}^2$. Fully controlled translation of the stage induces slow movement of the flake along a specific trajectory via spring forces, without rigidifying the sliding flake.

II. Translated energy density function

The initial energy density function $\rho_0(x', y', \lambda) = \eta\left[\cos\left(\frac{4\pi x'}{\sqrt{3}\lambda}\right) + 2\cos\left(\frac{2\pi x'}{\sqrt{3}\lambda}\right)\cos\left(\frac{2\pi y'}{\lambda}\right)\right]$ is defined in the moiré coordinate system, where y'-axis is aligned with moiré superlattice vector $\overrightarrow{R_M}$. It is essential to transform $\rho_0(x', y', \lambda)$ into the bottom layer coordinate system (with $\overrightarrow{R_b}$ as x-axis, as denoted in Figs. 1(c) and 1(f))) to obtain $\rho(x, y, \lambda)$, using a rotation matrix $\boldsymbol{R}(\beta)$. Here, β, the angle between $\overrightarrow{R_b}$ and $\overrightarrow{R_M}$, depends on θ and p (see Sec. IV).

A key step is to determine how $\rho(x, y, \lambda)$ evolves as the flake translates. Assuming the translation vector of the flake $\vec{v} = (\Delta x, \Delta y)$ decomposes as $x_1(\Delta x, \Delta y)\overrightarrow{R_{b1}} + x_2(\Delta x, \Delta y)\overrightarrow{R_{b2}}$, the moiré pattern undergoes translation as $\vec{V} = x_1(\Delta x, \Delta y)\overrightarrow{R_{M1}}(\theta, p) + x_2(\Delta x, \Delta y)\overrightarrow{R_{M2}}(\theta, p)$ [2]. Since $x_1(\Delta x, \Delta y), x_2(\Delta x, \Delta y)$ could be calculated, $\vec{V}$ can be expressed in terms of Δx and Δy. Given the moiré periodicity $\lambda(\theta, p) = \left|\overrightarrow{R_M}(\theta, p)\right| = \frac{p\left|\overrightarrow{R_b}\right|}{\sqrt{1+p^2-2p\cos\theta}}$, the translated energy density function simplifies to $\rho(x, y, \theta, p, \Delta x, \Delta y)$ in bottom layer coordinate system as

$$\rho(x, y, \theta, p, \Delta x, \Delta y) = 2\eta\cos\left[\frac{2\pi(py + p\Delta y - y\cos\theta + x\sin\theta)}{\sqrt{3}\left|\overrightarrow{R_b}\right|}\right]\cos\left[\frac{2\pi(-px - p\Delta x + x\cos\theta + y\sin\theta)}{\left|\overrightarrow{R_b}\right|}\right] + \eta\cos\left[\frac{4\pi(py + p\Delta y - y\cos\theta + x\sin\theta)}{\sqrt{3}\left|\overrightarrow{R_b}\right|}\right]. \quad \text{(S1)}$$

For homobilayers such as graphene on graphene system ($p = 1$), the translated energy density function is,

$$\rho(x, y, \theta, p = 1, \Delta x, \Delta y) = 2\eta\cos\left[\frac{2\pi(y + \Delta y - y\cos\theta + x\sin\theta)}{\sqrt{3}\left|\overrightarrow{R_b}\right|}\right]\cos\left[\frac{2\pi(-x - \Delta x + x\cos\theta + y\sin\theta)}{\left|\overrightarrow{R_b}\right|}\right] + \eta\cos\left[\frac{4\pi(y + \Delta y - y\cos\theta + x\sin\theta)}{\sqrt{3}\left|\overrightarrow{R_b}\right|}\right]. \quad \text{(S2)}$$

III. Structural frictionless theory for general parallelogram flake

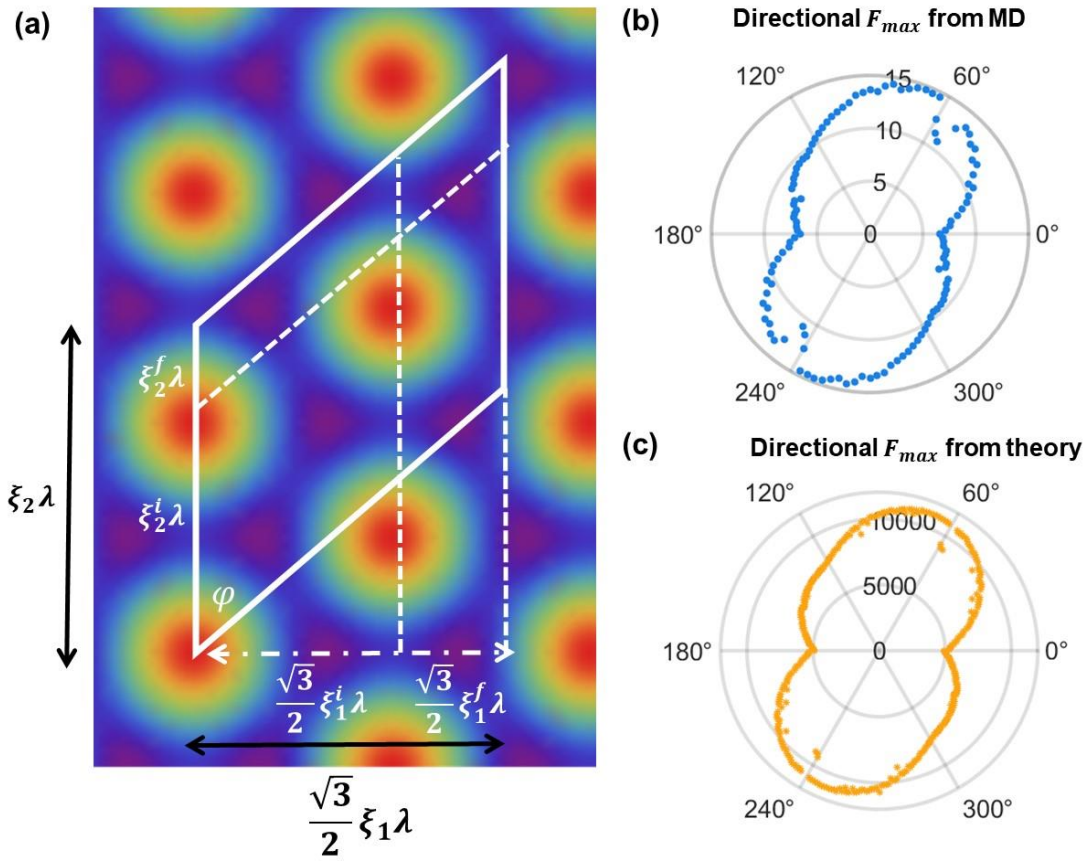

FIG.S2. (a) Sketch of the general parallelogram geometry. (b-c) Directional F_{max} for a parallelogram with $\xi_1 = 1.7, \xi_2 = 1.3, \varphi = 45°$: (b) from molecular dynamics simulation and (c) from Eq. (S4).

We specify the detailed configuration of the general parallelogram in Fig. S2(a). It corresponds to a general parallelogram with a base length of $\xi_2\lambda$ and a height of $\frac{\sqrt{3}}{2}\xi_1\lambda$, where the angle between its two sides is denoted as φ [Fig. S2(a)]. The parameters ξ_1 and ξ_2 can be decomposed into $\xi_{1,2} = \xi_{1,2}^i + \xi_{1,2}^f$ where $\xi_{1,2}^f$ representing the fractional components of $\xi_{1,2}$. The base of the parallelogram should pass through two six-fold axes of symmetry in the moiré pattern [Fig. S2(a)]. Integrating over the area of this general parallelogram yields the translational energy landscape $\Delta E(\xi_1, \xi_2, \Delta x, \Delta y, \varphi)$ as follows.

$$
\begin{aligned}
&\Delta E(\xi_1, \xi_2, \Delta x, \Delta y, \varphi) \\
&= \frac{\sqrt{3}\lambda^2\eta}{2\pi}\xi_2 \cos\left(\frac{4\pi\Delta y}{\sqrt{3}|\overrightarrow{R_b}|} + \pi\xi_1\right)\sin(\pi\xi_1) \\
&+ \frac{\sqrt{3}\lambda^2\eta}{2\pi^2(1+2\cos 2\varphi)}\left[4\cos\left(\frac{1}{6}\pi - \varphi\right)\sin\varphi \sin\left(\frac{\pi\xi_1(-1+\sqrt{3}\cot\varphi)}{2}\right)\cos\left(\frac{\pi(2\sqrt{3}\Delta x + 2\Delta y)}{\sqrt{3}|\overrightarrow{R_b}|}\right.\right. \\
&\left. + \frac{\pi(\xi_1 - 2\xi_2 - \sqrt{3}\xi_1\cot\varphi)}{2}\right)\sin(\pi\xi_2) \\
&+ 4\cos\left(\frac{1}{6}\pi + \varphi\right)\sin\varphi \sin\left(\frac{\pi\xi_1(1+\sqrt{3}\cot\varphi)}{2}\right)\cos\left(\frac{\pi(-2\sqrt{3}\Delta x + 2\Delta y)}{\sqrt{3}|\overrightarrow{R_b}|}\right. \\
&\left.\left. + \frac{\pi(\xi_1 + 2\xi_2 + \sqrt{3}\xi_1\cot\varphi)}{2}\right)\sin(\pi\xi_2)\right]. \qquad \text{(S3)}
\end{aligned}
$$

We can provide a more detailed analysis of Eq. (S3). When $\xi_2^f = 0$ for arbitrary φ, only the term containing $\xi_2 \cos\left(\frac{4\pi\Delta y}{\sqrt{3}|\overrightarrow{R_b}|} + \pi\xi_1\right)\sin(\pi\xi_1)$ remains. This results in directional frictionless interface for the parallelogram region with a base length of $\xi_2^i\lambda$ and a height of $\frac{\sqrt{3}}{2}\xi_1\lambda$. When $\xi_1^f = 0$ with $\varphi = \pi/3$, only the term involving $\xi_1 \cos\left(\frac{2\pi(\sqrt{3}\Delta x + \Delta y)}{\sqrt{3}|\overrightarrow{R_b}|} - \pi\xi_2\right)\sin(\pi\xi_2)$ persists, leading to directional frictionless interface for the parallelogram region with a base length of $\xi_1^i\lambda$ and a height of $\frac{\sqrt{3}}{2}\xi_2\lambda$. In fact, this case is equivalent to the scenario where $\xi_2^f = 0$ and $\varphi = \pi/3$, with the sole difference being a $120°$ rotation of the coordinate system. The condition $\varphi = \pi/3$ ensures that the base of the parallelogram passes through two six-fold axes of symmetry. Consequently, the vanishing-friction direction for the parallelogram is always aligned with $\overrightarrow{R_b}$.

From Eq. (S3), the friction force can be calculated with $f = -\partial\Delta E/\partial l$ for any sliding direction $l =< \cos\gamma, \sin\gamma >$, where γ is measured from the x-axis in bottom layer coordinate system.

$$
\begin{aligned}
f = {} & \frac{2\lambda^2\eta}{|\overrightarrow{R_b}|}\xi_2 \sin\gamma \sin\left(\frac{4\pi\Delta y}{\sqrt{3}|\overrightarrow{R_b}|} + \pi\xi_1\right)\sin(\pi\xi_1) \\
& + \frac{2\lambda^2\eta}{\pi|\overrightarrow{R_b}|(1+2\cos 2\varphi)}\Bigg\{2\sqrt{3}\cos\gamma\sin(\pi\xi_2)\sin\varphi\Bigg[\cos\left(\frac{1}{6}\pi \right. \\
& \left. - \varphi\right)\sin\left(\frac{\pi\xi_1(-1+\sqrt{3}\cot\varphi)}{2}\right)\sin\left(\frac{\pi(2\sqrt{3}\Delta x + 2\Delta y)}{\sqrt{3}|\overrightarrow{R_b}|} + \frac{\pi(\xi_1 - 2\xi_2 - \sqrt{3}\xi_1\cot\varphi)}{2}\right) \\
& - \cos\left(\frac{1}{6}\pi + \varphi\right)\sin\left(\frac{\pi\xi_1(1+\sqrt{3}\cot\varphi)}{2}\right)\sin\left(\frac{\pi(-2\sqrt{3}\Delta x + 2\Delta y)}{\sqrt{3}|\overrightarrow{R_b}|} + \frac{\pi(\xi_1 + 2\xi_2 + \sqrt{3}\xi_1\cot\varphi)}{2}\right)\Bigg] \\
& + 2\sin\gamma\sin(\pi\xi_2)\sin\varphi\Bigg[\cos\left(\frac{1}{6}\pi - \varphi\right)\sin\left(\frac{\pi\xi_1(-1+\sqrt{3}\cot\varphi)}{2}\right)\sin\left(\frac{\pi(2\sqrt{3}\Delta x + 2\Delta y)}{\sqrt{3}|\overrightarrow{R_b}|}\right. \\
& \left. + \frac{\pi(\xi_1 - 2\xi_2 - \sqrt{3}\xi_1\cot\varphi)}{2}\right) \\
& + \cos\left(\frac{1}{6}\pi + \varphi\right)\sin\left(\frac{\pi\xi_1(1+\sqrt{3}\cot\varphi)}{2}\right)\sin\left(\frac{\pi(-2\sqrt{3}\Delta x + 2\Delta y)}{\sqrt{3}|\overrightarrow{R_b}|}\right. \\
& \left. + \frac{\pi(\xi_1 + 2\xi_2 + \sqrt{3}\xi_1\cot\varphi)}{2}\right)\Bigg]\Bigg\}. \qquad \text{(S4)}
\end{aligned}
$$

For each sliding direction, we can obtain $F_{max} = \max(f)$ and compare it with simulation results. For example, we validate our theoretical formula by using a flake with $\xi_1 = 1.7$, $\xi_2 = 1.3$, and angle $\varphi = 45°$. The directional variation of F_{max} from MD simulations is shown in Fig. S2(b), while the directional friction predicted by Eq. (S4) is presented in Fig. S2(c) for comparison. Good agreement is observed between the theoretical friction results and MD simulations.

Next, we consider the effect of higher-order Fourier components in representing the moiré pattern, updating the energy density function as $\rho = \eta\sum_{n=1}^{\infty}\left[a_n\sum_{\vec{G}}\cos(n\vec{G}\cdot\vec{r})\right] = \sum_{n=1}^{\infty}a_n\eta\left[\cos\left(\frac{4\pi nx}{\sqrt{3}\lambda}\right) + 2\cos\left(\frac{2\pi nx}{\sqrt{3}\lambda}\right)\cos\left(\frac{2\pi ny}{\lambda}\right)\right]$. Here, a_n is a general coefficient accounting for the weight of $n\vec{G}$, and it is understood that a_n weakens significantly with increasing n. Integrating using the above series-form energy density function yields:

$$
\begin{aligned}
\Delta E = {} & \sum_{n=1}^{\infty} a_n\Bigg\{\frac{\sqrt{3}\lambda^2\eta}{2n\pi}\xi_2\cos\left(\frac{4n\pi\Delta y}{\sqrt{3}|\overrightarrow{R_b}|} + n\pi\xi_1\right)\sin(n\pi\xi_1) \\
& + \frac{\sqrt{3}\lambda^2\eta}{2n\pi}\Bigg[Z_1(n,\varphi,\xi_1)\cos\left(\frac{n\pi(2\sqrt{3}\Delta x + 2\Delta y)}{\sqrt{3}|\overrightarrow{R_b}|} + \frac{n\pi(\xi_1 - 2\xi_2 - \sqrt{3}\xi_1\cot\varphi)}{2}\right)\sin(n\pi\xi_2) \\
& + Z_2(n,\varphi,\xi_1)\cos\left(\frac{n\pi(-2\sqrt{3}\Delta x + 2\Delta y)}{\sqrt{3}|\overrightarrow{R_b}|}\right. \\
& \left. + \frac{n\pi(\xi_1 + 2\xi_2 + \sqrt{3}\xi_1\cot\varphi)}{2}\right)\sin(n\pi\xi_2)\Bigg]\Bigg\}, \qquad \text{(S5)}
\end{aligned}
$$

where $\zeta_{n1}(n,\varphi,\xi_1) = \frac{4\cos\left(\frac{1}{6}\pi-\varphi\right)\sin\varphi\sin\left(\frac{n\pi\xi_1(-1+\sqrt{3}\cot\varphi)}{2}\right)}{n\pi(1+2\cos 2\varphi)}$ and $\zeta_{n2}(n,\varphi,\xi_1) = \frac{4\cos\left(\frac{1}{6}\pi+\varphi\right)\sin\varphi\sin\left(\frac{n\pi\xi_1(1+\sqrt{3}\cot\varphi)}{2}\right)}{n\pi(1+2\cos 2\varphi)}$.

From Eq. (S5), for all-direction frictionless interface of the first kind, ξ_1 and ξ_2 must still be integers. For all-direction frictionless interface of the second kind, the condition $\xi_1^f = \zeta_{n1}(n, \varphi_m, \xi_1) = \zeta_{n2}(n, \varphi_m, \xi_1) = 0$ yields $\cot\varphi_m = \frac{\sqrt{3}}{3}\left(\frac{2m}{n\xi_1} + 1\right)$ where $m, n \in \mathbb{Z}^+$. Notably, the magic angle condition from the non-series-form energy density function—specifically $\cot\varphi_m = \frac{\sqrt{3}}{3}\left(\frac{2m}{\xi_1} + 1\right)$ as discussed in main text—remains a special case of $\cot\varphi_m = \frac{\sqrt{3}}{3}\left(\frac{2m}{n\xi_1} + 1\right)$ (with $n = 1$). Thus, the "magic angles" themselves remain unaffected.

For $\xi_2^f = 0$, we obtain the following equation in series form

$$\Delta E = \sum_{n=1}^{\infty} a_n \left[\frac{\sqrt{3}\eta\lambda^2}{2n\pi}\xi_2 \cos\left(n\pi\xi_1 + \frac{4n\pi\Delta y}{\sqrt{3}\,|\overrightarrow{R_b}|}\right)\sin(n\pi\xi_1)\right]. \tag{S6}$$

The frictional force for any sliding direction $l = (\cos\gamma, \sin\gamma)$, where γ is measured from the x-axis in bottom layer coordinate system is

$$f = \sum_{n=1}^{\infty} a_n \left[\frac{2\eta\lambda^2\xi_2 \sin\gamma \sin(n\pi\xi_1)\sin\left(n\pi\xi_1 + \frac{4n\pi\Delta y}{\sqrt{3}\,|\overrightarrow{R_b}|}\right)}{|\overrightarrow{R_b}|}\right]. \tag{S7}$$

The γ-dependent maximal frictional force $F_{max} = \max(f)$ is

$$F_{max} = \sum_{n=1}^{\infty} a_n \left(\frac{2\lambda^2\xi_2}{|\overrightarrow{R_b}|}\sin(n\pi\xi_1)\sin\gamma\right). \tag{S8}$$

From Eq. (S7), a directional frictionless interface requires $\sin\gamma \sin\left(n\pi\xi_1 + \frac{4n\pi\Delta y}{\sqrt{3}\,|\overrightarrow{R_b}|}\right) = 0$ when ξ_1 is not an integer. This condition is satisfied if either $\sin\gamma = 0$ or $\sin\left(n\pi\xi_1 + \frac{4n\pi\Delta y}{\sqrt{3}\,|\overrightarrow{R_b}|}\right) = 0$. The case $\sin\gamma = 0$ indicates that the frictionless direction is along x direction (i.e., Δy remains constant during sliding), which corresponds to the direction of the bottom layer lattice vector $\overrightarrow{R_b}$. However, From Eq. (S6), the condition $\eta\cos\left(n\pi\xi_1 + \frac{4n\pi\Delta y}{\sqrt{3}\,|\overrightarrow{R_b}|}\right) = -1$ (i.e., equivalent to $\sin\left(n\pi\xi_1 + \frac{4n\pi\Delta y}{\sqrt{3}\,|\overrightarrow{R_b}|}\right) = 0$), is required to ensure that directional frictionless motion occurs along local energy minima path. Notably, the frictionless condition derived from the non-series-form energy density function (i.e., $\sin\left(\xi_1\pi + \frac{4\pi\Delta y}{\sqrt{3}\,|\overrightarrow{R_b}|}\right) = 0$ discussed in main text) also applies in this context.

IV. Visualizing β

For a directional frictionless interface, the vanishing-friction direction can be obtained by an anticlockwise rotation β from the base side, with its formula given in Eq. (S9) (in radians). We also present a 3D plot of β as a function of the relative twist angle θ and lattice constant ratio p in Fig. S3. The green line corresponds to the homobilayer case with $p = 1$ (e.g., graphene on graphene homobilayers). It is also noted that β represents the angle between bottom

layer lattice vector $\overrightarrow{R_b}$ and moiré lattice vectors $\overrightarrow{R_M}$.

$$\beta(\theta,p)=\begin{cases}\dfrac{\pi+\theta}{2} & (p=1)\\ \arccos\left(\dfrac{\cos\theta-p}{\sqrt{1+p^2-2p\cos\theta}}\right) & (p\neq 1)\end{cases}. \tag{S9}$$

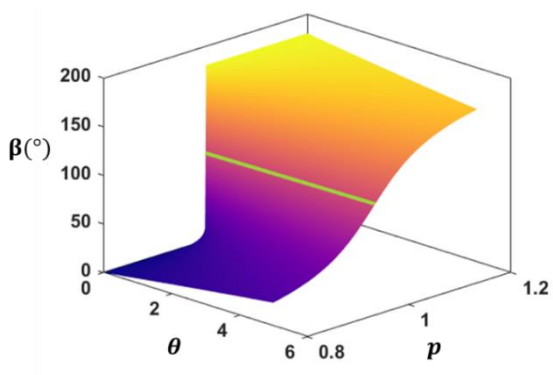

FIG. S3. Visualization of β as a function of relative twist angle θ and lattice constant ratio p. The green line denotes $\beta(\theta)$ for $p=1$.

V. General parallelogram flake: All-direction frictionless of 1st kind

From Eq. (S3), when ξ_1 and ξ_2 are integers ($\xi_1^f=\xi_2^f=0$), the flake is all-direction frictionless of the first kind. In this scenario, the flake area is a multiple of a complete moiré tile, consistent with previous studies [6,14–20]. Notably, all-direction frictionless interface of the first kind is independent of the angle φ. Thus, beyond the most common rhombic moiré supercell with $\varphi=60°$, other non-rhombic configurations can also be all-direction frictionless.

Fig. S4(a) shows directional F_{max} for the first kind of all-direction frictionless graphene on graphene system across different φ, compared with results for directional frictionless system [i.e., Fig. 1(b) in main text]. Fig. S4(b) presents analogous results for the graphene on h-BN system [i.e., Fig. 1(e) in main text]. Fig. S4(c) illustrates the all-direction frictionless moiré patterns within flake with varying φ in the graphene on graphene system, where the base length equals λ and the height equals $\frac{\sqrt{3}}{2}\lambda$ (i.e., $\xi_1=\xi_2=1$).

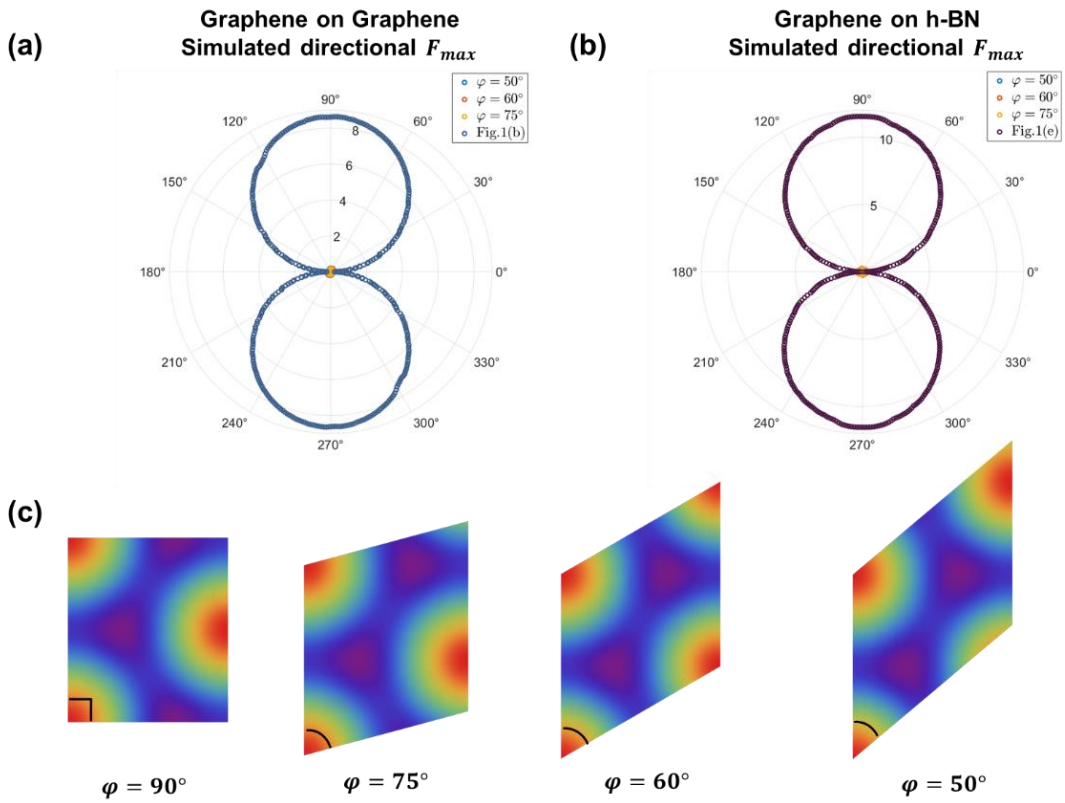

FIG. S4. Results for all-direction vanishing-friction of 1st kind. (a-b) Comparison of directional F_{max} across different φ with results from Figs. 1(b) and 1(e) (directional frictionless). (c) Eligible moiré patterns for varying φ.

VI. General parallelogram flake: All-direction frictionless of 2nd kind

According to the main text, all-direction frictionless interface of 2nd kind requires $\xi_1^f = 0$ and $\zeta_1(\varphi, \xi_1^i) = \zeta_2(\varphi, \xi_1^i) = 0$,

where $\zeta_1(\varphi, \xi_1) = \frac{4\cos\left(\frac{1}{6}\pi - \varphi\right)\sin\varphi\sin\left(\frac{\pi\xi_1(-1+\sqrt{3}\cot\varphi)}{2}\right)}{\pi(1+2\cos 2\varphi)}$, and $\zeta_2(\varphi, \xi_1) = \frac{4\cos\left(\frac{1}{6}\pi + \varphi\right)\sin\varphi\sin\left(\frac{\pi\xi_1(1+\sqrt{3}\cot\varphi)}{2}\right)}{\pi(1+2\cos 2\phi)}$.

Here we require both $\sin\left(\frac{\pi\xi_1(-1+\sqrt{3}\cot\varphi_m)}{2}\right) = 0$ and $\sin\left(\frac{\pi\xi_1(1+\sqrt{3}\cot\varphi_m)}{2}\right) = 0$ to hold simultaneously for some φ_m. The solution is

$$\cot\varphi_m = \frac{\sqrt{3}}{3}\left(1 + \frac{2m}{\xi_1}\right), \tag{S10}$$

with m, ξ_1 are positive integers.

To explain simulation results in Figure 1 of the main text, the parameters $\xi_1 = 1, \varphi = \operatorname{arccot}\frac{5}{\sqrt{3}}$ $(= 19.1°)$, and $\xi_1 = 2, \varphi = \operatorname{arccot}\frac{2}{\sqrt{3}}$ $(= 40.89°)$, are indeed magic-angle flake parameters.

Since there are no specific constraints on ξ_2, even flakes whose area is not an integer multiple of the moiré tile can be all-direction frictionless.

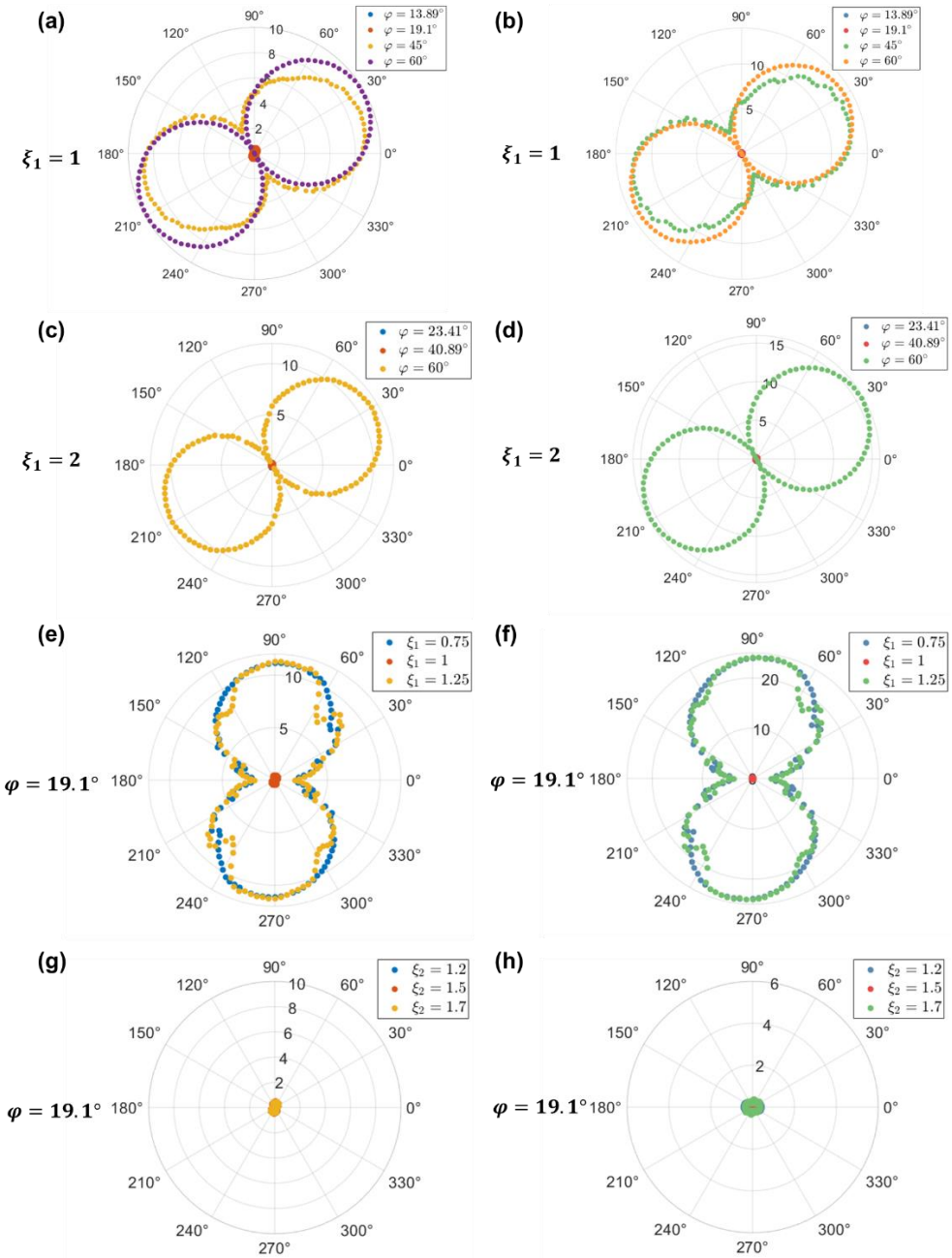

FIG. S5. Representative molecular simulation results for all-direction vanishing-friction of 2nd kind. (a) For $\xi_1 = 1$, $\xi_2 = 1.5$, directional F_{max} results across different φ (13.89° and 19.1° are magic angles). (c) For $\xi_1 = 2$, $\xi_2 = 1.5$, directional F_{max} results across different φ (23.41° and 40.89° are magic angles). (e) For $\varphi = 19.1°, \xi_2 = 1.5$, directional F_{max} results across different ξ_1. (g) For $\varphi = 19.1°, \xi_1 = 1$, directional F_{max} results across different ξ_2. Panels (a), (c), (e), and (g) correspond to the graphene on graphene system, while (b), (d), (f), and (h) show the corresponding results for the graphene on h-BN system.

Molecular dynamics simulation of graphene on graphene and graphene on h-BN systems are performed to validate additional magic angle results. In Figs. S5(a), S5(b), simulated directional F_{max} is plotted for $\xi_1 = 1, \xi_2 = 1.5$ and varying φ. Friction approaches zero in all directions when φ reaches magic angles of 19.1° and 13.89° ($= \operatorname{arccot}\frac{7}{\sqrt{3}}$). For $\xi_1 = 2, \xi_2 = 1.5$ [Figs. S5(c) and S5(d)], the magic angles 40.89° and 23.41°($= \operatorname{arccot}\frac{4}{\sqrt{3}}$) also align with predictions. Notably, at $\varphi = 60°$, $\lim_{\varphi \to \pi/3} \zeta_1(\varphi, \xi_1^i) = \xi_1^i$ and $\lim_{\varphi \to \pi/3} \zeta_2(\varphi, \xi_1^i) = \sin(\pi \xi_1^i)/\pi = 0$. This

leaves only the second term of Eq. (S3) intact, resulting in directional frictionless interface along the $120°$ direction.

Figs. S5(e), S5(f) show results for parallelograms with $\xi_2 = 1.5, \varphi = 19.1°$, and varying ξ_1, confirming that all-direction vanishing-friction occurs exclusively when ξ_1 is an integer. Additionally, Figs. S5(g), S5(h) demonstrate that ξ_2 has a negligible effect on perturbing all-direction vanishing-friction of the second kind.

VII. Structural frictionless theory for regular hexagon flake

This section examines regular hexagon flakes with a side length of $\xi\lambda$. We assume that the center of the regular hexagon lies at a six-fold axis of symmetry in the moiré pattern, with the moiré vector forming an angle ϕ with the circumradius of the hexagon [Fig. S6(a)]. Integrating over its area yields the translational energy landscape as

$$\Delta E = 9\eta\lambda^2 Z(\phi,\xi)\left[\cos\left(\frac{4\Delta y\pi}{\sqrt{3}|\overrightarrow{R_b}|}\right) + \cos\left(\frac{2(3\Delta x - \sqrt{3}\Delta y)\pi}{3|\overrightarrow{R_b}|}\right) + \cos\left(\frac{2(3\Delta x + \sqrt{3}\Delta y)\pi}{3|\overrightarrow{R_b}|}\right)\right], \tag{S11}$$

where

$$\begin{aligned} Z(\phi,\xi) = -\frac{1}{4\pi^2(1+2\cos(2\phi))(\sqrt{3}-3\cot\phi)}\Bigg\{&\cos(2\pi\xi\cos\phi)\cos\left(\frac{2\pi\xi\sin\phi}{\sqrt{3}}\right) \\ &+\cos\left(\frac{4\pi\xi\sin\phi}{\sqrt{3}}\right)(-1+\sqrt{3}\cot\phi) \\ &+\cot\phi\left[-\sqrt{3}\cos\left(2\pi\xi\cos\phi - \frac{2\pi\xi\sin\phi}{\sqrt{3}}\right) + 3\cot\phi\sin(2\pi\xi\cos\phi)\sin\left(\frac{2\pi\xi\sin\phi}{\sqrt{3}}\right)\right]\Bigg\}. \end{aligned}$$

The only condition for achieving frictionless interface is to vanish $Z(\phi,\xi)$. For demonstration, we consider two special cases where $\phi = 0°$ and $\phi = 15°$.

$$Z(\phi = 0°,\xi) = \frac{\sin(\pi\xi)}{6\sqrt{3}\pi^2}[2\pi\xi\cos(\pi\xi) + \sin(\pi\xi)], \tag{S12}$$

$$Z(\phi = 15°,\xi) = \frac{\sqrt{3}\cos\left(2\sqrt{\frac{2}{3}}\pi\xi\right) - \sqrt{3}\cos\left(\sqrt{\frac{2}{3}}\pi\xi\right)\cos(\sqrt{2}\pi\xi) + 3\sin\left(\sqrt{\frac{2}{3}}\pi\xi\right)\sin(\sqrt{2}\pi\xi)}{12\pi^2}. \tag{S13}$$

Flakes with $\phi = 0$ and $\phi = 15°$ are shown in Fig. S6(b) in the substrate coordinate system of graphene on graphene (with $\xi = 1$ for demonstration). The numerical solution to $Z(\phi = 0°,\xi) = 0$ yields $\xi_m = 0.584$, 1, $1.533, \ldots$, while solving $Z(\phi = 15°,\xi) = 0$ gives $\xi_m = 0.581$, 1.064, $1.531, \ldots$.

We find that for certain "magic sizes" ξ_m (i.e., non-integer solutions), the hexagon can still be all-direction frictionless even when its area is not an integer multiple of a complete moiré tile. Molecular dynamics simulations have been performed to validate these results in graphene on graphene and graphene on h-BN system [Fig. S6(c), S6(d)], showing that the magic size ξ_m causes a sharp reduction in F_{max}. For other values of ξ (e.g., $\xi = 0.8$), $Z(\phi,\xi)$ could not be eliminated, resulting in significantly larger friction for the flake.

It should be noted that for Eq. (S11) to hold, the center of the regular hexagon must lie at a six-fold axis of symmetry in the moiré pattern. However, the method described in Sec. IX can be used to calculate energy function of a regular

hexagon for other initially-translated configurations.

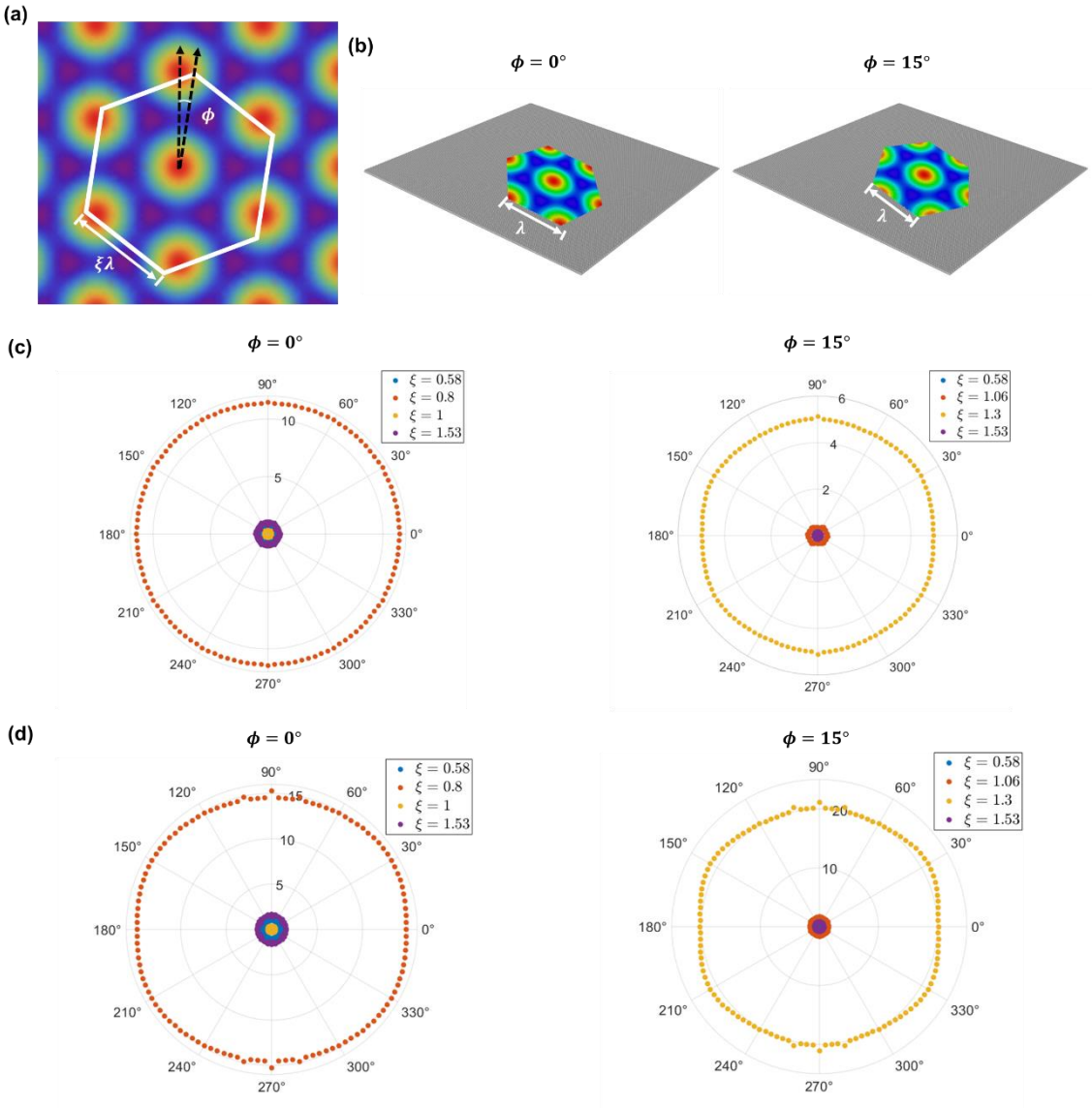

FIG. S6. (a) Schematic of a regular hexagon flake with side length $\xi\lambda$ and angle φ for graphene on graphene interface, illustrating the moiré pattern. (b) Regular hexagon flakes with $\xi = 1$ for $\phi = 0°$ and $\phi = 15°$. (c) Directional F_{max} from molecular dynamics (MD) simulations for graphene on graphene interface. (d) Directional F_{max} from MD simulation for graphene on h-BN interface.

VIII. Structural frictionless theory for equilateral triangle flake

This section examines equilateral triangle flake with a side length of $\xi\lambda$. We assume one vertex lies at a six-fold axis of symmetry in the moiré pattern, with the moiré vector forming an angle ϕ with one side [Fig. S7(a)]. Flakes with $\phi = 0°$ and $\phi = 30°$ for graphene on graphene interface are shown in Fig. S7(b), where $\xi = 1$ for $\phi = 0°$ and $\xi = \sqrt{3}$ for $\phi = 30°$. The analytical scaling expressions of the translational interface energy are given as,

$$\Delta E(\phi = 0°) = \frac{\sqrt{3}\eta\lambda^2}{4\pi^2}\left\{-\pi\xi \sin\left(\frac{4\pi\Delta y}{\sqrt{3}|\overrightarrow{R_b}|}\right) + \sin(\pi\xi)\sin\left(\frac{4\pi\Delta y}{\sqrt{3}|\overrightarrow{R_b}|} + \pi\xi\right) + \cos\left(\frac{2\pi\Delta x}{|\overrightarrow{R_b}|} - \pi\xi\right)\left[-2\sin\left(\frac{2\pi\Delta y}{\sqrt{3}|\overrightarrow{R_b}|}\right)\sin(\pi\xi) + 2\pi\xi\sin\left(\frac{2\pi\Delta y}{\sqrt{3}|\overrightarrow{R_b}|} + \pi\xi\right)\right]\right\}, \tag{S14}$$

$$\Delta E(\phi = 30°) = \frac{3\sqrt{3}\eta\lambda^2}{4\pi^2}\left[\cos\left(\frac{4\pi\Delta y}{\sqrt{3}|\overrightarrow{R_b}|}\right) + \cos\left(\frac{2\pi\left(\sqrt{3}\Delta x + \Delta y - |\overrightarrow{R_b}|\xi\right)}{\sqrt{3}|\overrightarrow{R_b}|}\right) + \cos\left(\frac{2\pi\left(-\sqrt{3}\Delta x + \Delta y + |\overrightarrow{R_b}|\xi\right)}{\sqrt{3}|\overrightarrow{R_b}|}\right)\right]\sin^2\left(\frac{\pi\xi}{\sqrt{3}}\right). \tag{S15}$$

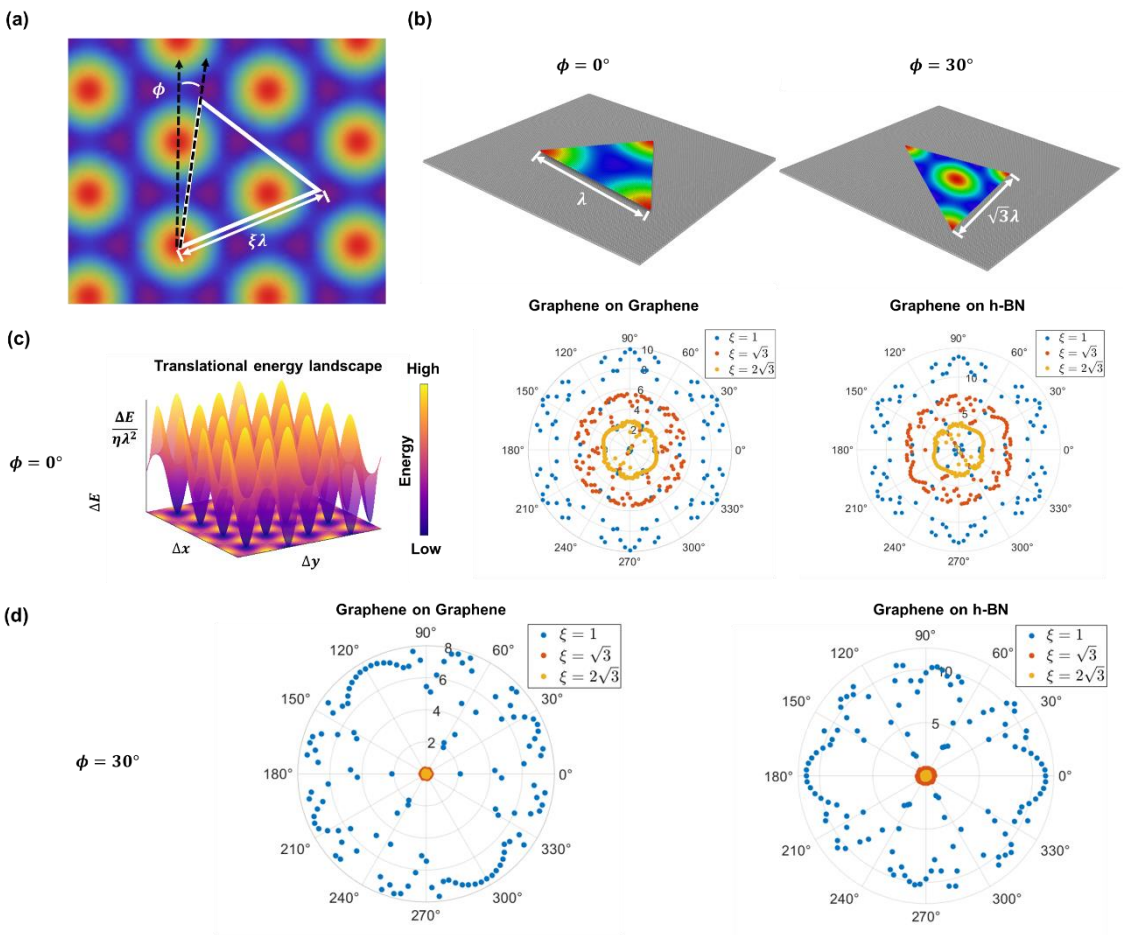

FIG. S7. (a) Schematic illustration of the moiré pattern for an equilateral triangle flake with side length $\xi\lambda$ and angle ϕ for graphene on graphene interface. (b) Equilateral triangle flakes with $\xi = 1$ (for $\phi = 0°$) and $\xi = \sqrt{3}$ (for $\phi = 30°$). (c) Translational energy landscape ($\xi = 1$) and directional F_{max} from molecular dynamics (MD) simulation for $\phi = 0°$. (d) Directional F_{max} from MD simulations for $\phi = 30°$.

From Eq. (S14), for $\phi = 0°$, ΔE cannot vanish, making a frictionless state impossible. The normalized translational energy landscape for $\xi = 1$ is shown on the left of Fig. S7(c). It should be noted that the directions where

$\partial\Delta E/\partial l = 0$ are not frictionless, as there is no continuous zero-gradient valley path. Since the translational energy landscape described by Eq. (S14) is analogous to that in Fig. S7(c) for any ξ, we conclude that equilateral triangular flakes with $\phi = 0°$ cannot exhibit frictionless properties under these conditions.

From Eq. (S15), for $\phi = 30°$, "magic sizes" given by $\xi = \sqrt{3}n$ $(n \in \mathbb{Z}^+)$ cause ΔE to vanish, enabling all-direction frictionless interface. At these magic sizes, the area of the equilateral triangle flake is not an integer multiple of a complete moiré tile, an observation similar to that for regular hexagons. MD simulations for graphene on graphene and graphene on h-BN systems [Fig. S7(c), S7(d)] support these conclusions.

Further numerical investigations show that frictionless properties cannot be achieved for $\phi \neq 30°$.

It should be noted that for Eq. (S14), Eq. (S15) to hold, one vertex of the triangle must lie at a six-fold axis of symmetry in the moiré pattern. However, the method described in Sec. IX can be used to calculate energy function of triangles for other initially-translated configurations.

IX. Assemble frictionless flakes as "Lego Brick"

It is possible to assemble frictionless elementary flake shapes (e.g., parallelograms, equilateral triangles, regular hexagons) to construct more complex flake geometries. Consider the white-line-boundary parallelogram flake Ω_{i} with its initial moiré distribution shown in Fig. S8(a). It should be noted that Eq. (S3) is not directly applicable to Ω_{i}, as Eq. (S3) is derived for the standard initial moiré distribution (i.e., with one vertex of the parallelogram at a six-fold axis of symmetry) in the yellow-dashed-boundary parallelogram $\Omega_i{}'$. However, the required energy function can be obtained by transforming Eq. (S3) as follows.

We identify a shifting vector $(\delta x_i, \delta y_i)$ such that Ω_{i} and $\Omega_{\mathrm{i}}{}'$(yellow dashed line) are connected by the black arrow $(\delta x_i, \delta y_i)$ within the moiré background of Ω_{i}. Importantly, $(\delta x_i, \delta y_i)$ arises from the translation of $\Omega_i{}'$ by (dx_i, dy_i) to Ω_i in bottom layer coordinate system [Fig. S8(b)]. From moiré geometry, the relationship is given by,

$$\begin{cases} dx_i = \dfrac{\delta y_i \left|\overrightarrow{R_b}\right|}{\lambda} \\ dy_i = -\dfrac{\delta x_i \left|\overrightarrow{R_b}\right|}{\lambda} \end{cases}, \tag{S16}$$

where $\left|\overrightarrow{R_b}\right|$ is the lattice constant of the bottom layer and λ is moiré wavelength.

In this manner, we obtain the energy function for individual shapes as $\Delta E(\xi_1, \xi_2, dx_i + \Delta x, dy_i + \Delta y, \varphi)$. Ultimately, by summing these contributions, we can calculate the translational energy function for the entire assembled flake.

For example, consider the assembled flake $\Omega = \Omega_1 + \Omega_2$ in Fig. S8(c). Their parameters are: Ω_1 $(\xi_1 = \frac{4}{3}, \xi_2 = 1,\ \varphi = \frac{\pi}{3})$ and Ω_2 $(\xi_1 = \frac{4}{3}, \xi_2 = 1, \varphi = \frac{2\pi}{3})$. The standard result for Ω_1 is $\Delta E_1(\xi_1, \xi_2, \Delta x, \Delta y, \varphi) =$

$\Delta E\left(\frac{4}{3}, 1, \Delta x, \Delta y, \frac{\pi}{3}\right)$ (where ΔE is given by Eq. (S3)). From the moiré map, we read $(\delta x_2, \delta y_2) = \left(\frac{\sqrt{3}\lambda}{6}, \frac{\lambda}{6}\right)$, and since $\lambda = \frac{|\overrightarrow{R_b}|}{2\sin\frac{\theta}{2}}$, Eq. (S16) yields $(dx_2, dy_2) = (\frac{|\overrightarrow{R_b}|}{6}, -\frac{|\overrightarrow{R_b}|}{2\sqrt{3}})$. The energy function for Ω_2 is thus $\Delta E_2(\xi_1, \xi_2, \Delta x, \Delta y, \varphi) = \Delta E\left(\frac{4}{3}, 1, \frac{|\overrightarrow{R_b}|}{6} + \Delta x, -\frac{|\overrightarrow{R_b}|}{2\sqrt{3}} + \Delta y, \frac{2\pi}{3}\right)$. Finally, the energy function for the assembled shape Ω is

$$\Delta E_\Omega = \Delta E_1 + \Delta E_2. \tag{S17}$$

From Eq. (S17), we can derive the directional F_{max} of flake Ω, shown in Fig. S8(d). The results are in good agreement with simulation data. Since the two parallelograms exhibit frictionless properties along the same direction, the assembled flake is also directional frictionless.

This analysis could be readily extended to other scenarios. Other building blocks—such as regular hexagonal flakes (Sec. VII) and equilateral triangular flake (Sec. VIII)—can also be utilized to "assemble" complex geometries, and their energy landscapes can be analytically derived. The key step is identifying the shifting vector $(\delta x_i, \delta y_i)$ relative to the standard moiré pattern.

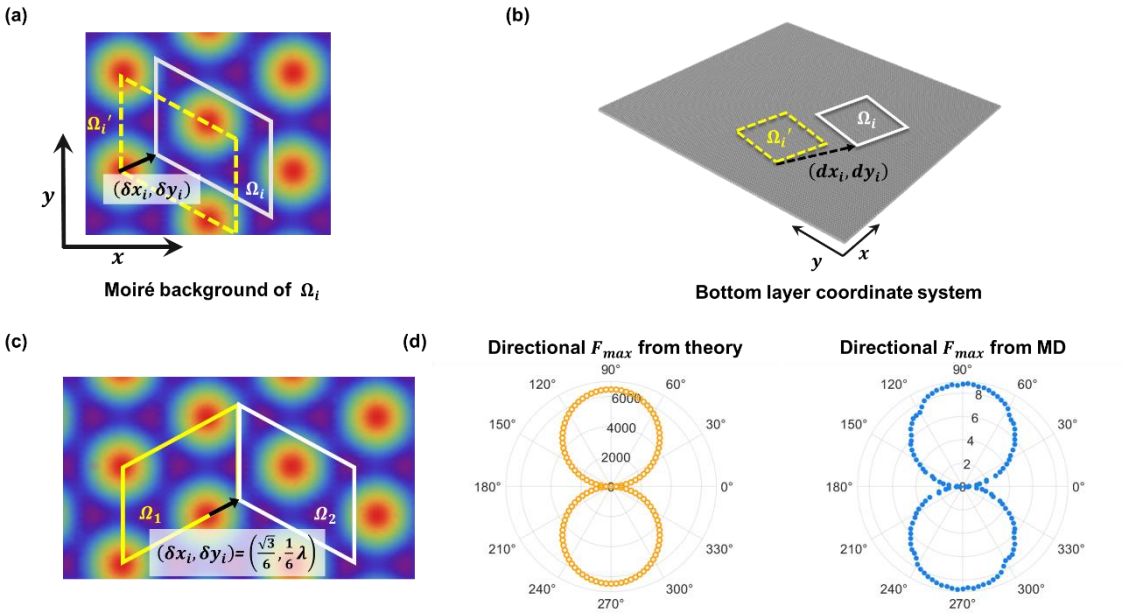

FIG. S8. (a) The parallelogram Ω_i (white solid line) and its standard counterpart Ω_i' (yellow dashed line) are connected by black arrowed shift vector $(\delta x_i, \delta y_i)$ within the moiré background of Ω_i. (b) The shift vector $(\delta x_i, \delta y_i)$ in (a) arises from the translation of Ω_i' to Ω_i via the vector (dx_i, dy_i) in the bottom layer coordinate system. (c) A composite shape assembled from two parallelograms. (d) Comparison of the directional friction results from our theoretical method and molecular dynamics (MD) simulations.

X. Enhanced rotational stability

Large flakes containing multiple complete and incomplete moiré tiles exhibit enhanced rotational stability. We consider a parallelogram flake with $\varphi = 60°$ and $\xi_1 = \xi_2 + 0.5$ [Fig. S9(a)], where complete moiré tiles are shown in white and the incomplete ones in the yellow region on the right. Molecular dynamics simulations are performed on flakes with varying ξ_2 to investigate rotational stability. The flake is unconstrained relaxed for 30 ps in an NVT

canonical ensemble at 0.1K. The twist angle is calculated as the average of the twist angles of all flake atoms relative to the initial positions. As demonstrated in Fig. S9(b), increasing ξ_2 reduces the deviation of the twist angle from its initial value, with the change becoming negligible ($< 0.1°$) when $\xi_2 = 50$.

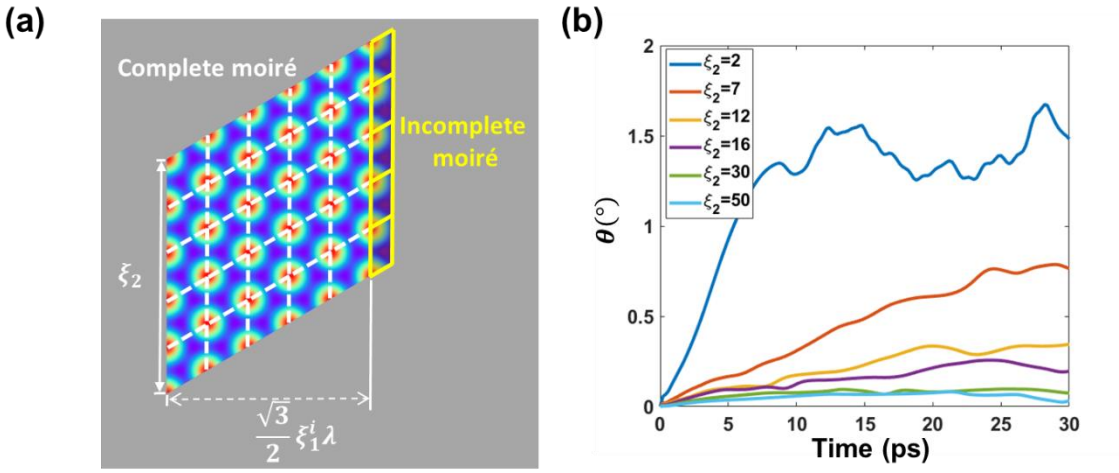

FIG. S9. (a) Parallelogram flake composed of complete moiré supercells (left, white) with $\xi_2 = 5, \xi_1 = 5.5$ and incomplete moiré tiles (right, yellow). (b) Deviation of the twist angle from its initial value decreases with increasing ξ_2.

XI. Relative displacement vectors to tune frictionless state in nanoconfinement channels

TABLE I. Example vectors $\vec{d} = (x_0, y_0)$

Flake shape	**Single-interface property**	**Double-interface property**	(x_0, y_0) **in units of** $\lvert\overrightarrow{R_{bot}}\rvert$
Parallelogram	Directional frictionless (along $0°$)	All-direction frictionless	$\left(0, \frac{\sqrt{3}}{4}\right)$
Equilateral triangle	Non-vanishing friction	Directional frictionless (along $0°$)	$\left(0, \frac{\sqrt{3}}{2}\right)$ or $\left(\frac{1}{2}, 0\right)$
Equilateral triangle	Non-vanishing friction	Directional frictionless (along $60°$)	$\left(\frac{1}{4}, \frac{\sqrt{3}}{4}\right)$
Equilateral triangle	Non-vanishing friction	Directional frictionless (along $-60°$)	$\left(\frac{1}{4}, \frac{3\sqrt{3}}{4}\right)$

For bilayer nanoconfinement channels, the relative displacement of the channel's top layer with respect to the bottom layer can be used to tune friction behaviors, as shown in Table I. The parallelogram is defined with $\xi_2 = 1$ and $\xi_1 = 2/3$. For the equilateral triangle, the parameters are $\phi = 0°$ and a side length of λ. We denote the relative displacement of the channel's top layer with respect to the bottom layer as the vector $\vec{d} = (x_0, y_0)$, with units of $\lvert\overrightarrow{R_{bot}}\rvert$. Results are presented in the coordinate system of the channel's bottom layer, where the x-axis is aligned with the bottom layer's lattice vector $\overrightarrow{R_{bot}}$. AA stacking corresponds to $\vec{d} = (0, 0)$.

XII. Translational interface energy of rectangular graphene with respect to bottom and top h-BN layer

For graphene nanoribbons embedded in h-BN channels, the x-axis is aligned with the zigzag direction of AA' stacked h-BN channel layers [21], which is also parallel to the lattice vector of the channel's bottom layer, $\overrightarrow{R_{bot}}$.The relative twist between layers is zero. Thus, the initial energy density function of graphene flake with respect to the bottom h-BN layer is $\rho_{bot0}(x,y,\lambda) = -\cos\left(\frac{4\pi y}{\sqrt{3}\lambda}\right) - 2\cos\left(\frac{2\pi y}{\sqrt{3}\lambda}\right)\cos\left(\frac{2\pi x}{\lambda}\right)$. For a translation vector $(\Delta x, \Delta y)$ of the graphene layer, the translated energy density function is $\rho_{bot}(x,y,\Delta x,\Delta y,\lambda) = \rho_{bot0}\left(x - \frac{\Delta x\lambda}{|\overrightarrow{R_{bot}}|}, y - \frac{\Delta y\lambda}{|\overrightarrow{R_{bot}}|}, \lambda\right)$. For a rectangular graphene nanoribbon with length $2a\lambda$ and width $2b\lambda$ (λ=15 nm), the translational interface energy with respect to bottom h-BN layer is

$$\begin{aligned}\Delta E_{bot} &= \int_{-a\lambda}^{a\lambda}\int_{-b\lambda}^{b\lambda} \rho_{bot}(x,y,\Delta x,\Delta y,\lambda)\, dydx \\ &= -\frac{\sqrt{3}\lambda^2}{\pi^2}\left[2\cos\left(\frac{2\pi\Delta x}{|\overrightarrow{R_{bot}}|}\right)\cos\left(\frac{2\pi\Delta y}{\sqrt{3}|\overrightarrow{R_{bot}}|}\right)\sin(2a\pi)\sin\left(\frac{2b\pi}{\sqrt{3}}\right) + a\pi\cos\left(\frac{4\pi\Delta y}{\sqrt{3}|\overrightarrow{R_{bot}}|}\right)\sin\left(\frac{4b\pi}{\sqrt{3}}\right)\right].\end{aligned} \tag{S18}$$

The top h-BN layer is AA' stacked with bottom h-BN layer. This configuration is mathematically equivalent to first rotating $\rho_{bot}(x,y,\Delta x,\Delta y,\lambda)$ by 180° in the $x-y$ plane and then translating it downward by $\frac{1}{\sqrt{3}}\lambda$. Through this manipulation, the initial energy density function for the top h-BN layer is obtained as $\rho_{top0}(x,y,\lambda) = -\cos\left(\frac{4\pi+4\sqrt{3}\pi y}{3\lambda}\right) - 2\cos\left(\frac{2\pi x}{\lambda}\right)\cos\left(\frac{2\pi+2\sqrt{3}\pi y}{3\lambda}\right)$. Then, the translated energy density function is $\rho_{top}(x,y,\Delta x,\Delta y,\lambda) = \rho_{top0}\left(x - \frac{\Delta x\lambda}{|\overrightarrow{R_{bot}}|}, y - \frac{\Delta y\lambda}{|\overrightarrow{R_{bot}}|}, \lambda\right)$. After integration, the translation interface energy with respect to top h-BN layer is

$$\begin{aligned}\Delta E_{top} &= \int_{-a\lambda}^{a\lambda}\int_{-b\lambda}^{b\lambda} \rho_{top}(x,y,\Delta x,\Delta y,\lambda)\, dydx \\ &= -\frac{\sqrt{3}\lambda^2}{\pi^2}\left\{2\cos\left(\frac{2\pi\Delta x}{|\overrightarrow{R_{bot}}|}\right)\cos\left[\frac{2}{3}\pi\left(\frac{\sqrt{3}\Delta y}{|\overrightarrow{R_{bot}}|} - \frac{1}{\lambda}\right)\right]\sin(2a\pi)\sin\left(\frac{2b\pi}{\sqrt{3}}\right) + a\pi\cos\left[\frac{4}{3}\pi\left(\frac{\sqrt{3}\Delta y}{|\overrightarrow{R_{bot}}|} - \frac{1}{\lambda}\right)\right]\sin\left(\frac{4b\pi}{\sqrt{3}}\right)\right\}.\end{aligned} \tag{S19}$$

Summing Eq. (S18) and Eq. (S19) gives the total energy function,

$$\begin{aligned}\Delta E_{total} = -\frac{2\sqrt{3}\lambda^2}{\pi^2}[&2\cos\left(\frac{2\pi\Delta x}{|\overrightarrow{R_{bot}}|}\right)\cos\left(\frac{2\Delta y\pi}{\sqrt{3}|\overrightarrow{R_{bot}}|} - \frac{\pi}{3\lambda}\right)\cos\left(\frac{\pi}{3\lambda}\right)\sin(2a\pi)\sin\left(\frac{2b\pi}{\sqrt{3}}\right) \\ &+a\pi\cos\left(\frac{4\Delta y\pi}{\sqrt{3}|\overrightarrow{R_{bot}}|} - \frac{2\pi}{3\lambda}\right)\cos\left(\frac{2\pi}{3\lambda}\right)\sin\left(\frac{4b\pi}{\sqrt{3}}\right)].\end{aligned} \tag{S20}$$

Continuous zero-gradient valley paths emerge along the x direction when the length parameter satisfies $a = \frac{n}{2}$ (where n is an integer). This direction aligns with both the zigzag direction of h-BN channels and the moiré translation direction of graphene (since $\beta = 0°$). Such length values, enabling directional vanishing friction, facilitate growth along

the zigzag direction of h-BN channels. Meanwhile, energy barriers between adjacent zero-gradient valley paths restrict the nanoribbon width from increasing significantly. Consequently, ultra-high-aspect-ratio graphene nanoribbons are observed to grow in h-BN channels.

XIII. Translational interface energy of strain-engineered rectangular graphene on graphene system

In this section, we establish the relationship between strain and frictional force for tensile-strained rectangular graphene flakes. The top-layer graphene is a square with side length L_0 and is initially aligned with the bottom layer in an AA stacking configuration.

We first discuss biaxial strain ε (>0) applied on the flake [Fig. S10(a)]. In this case, the flake length becomes $L_0(1+\varepsilon)$, and the moiré wavelength changes to $\lambda(\varepsilon)=|\overrightarrow{R_b}|\left(1+\frac{1}{\varepsilon}\right)$. By substituting $\xi_1=\frac{L_0(1+\varepsilon)}{\frac{\sqrt{3}}{2}\lambda(\varepsilon)}, \xi_2=\frac{L_0(1+\varepsilon)}{\lambda(\varepsilon)}, \varphi=\frac{\pi}{2}$ into Eq. (S3), we can rewrite the translational energy landscape for the square graphene flake as

$$\Delta E_{biaxial}(\Delta x,\Delta y,\varepsilon,L_0) = \frac{\sqrt{3}|\overrightarrow{R_b}|(1+\varepsilon)^2}{2\pi^2\varepsilon^2}\left[2|\overrightarrow{R_b}|\left(\cos\left(\frac{\pi\left(6\Delta x+2\sqrt{3}\Delta y+\left(-3+\sqrt{3}\right)L_0\varepsilon\right)}{3|\overrightarrow{R_b}|}\right)\right.\right.$$
$$+\cos\left(\frac{\pi\left(-6\Delta x+2\sqrt{3}\Delta y+\left(3+\sqrt{3}\right)L_0\varepsilon\right)}{3|\overrightarrow{R_b}|}\right)\Bigg)\sin\frac{L_0\pi\varepsilon}{|\overrightarrow{R_b}|}\sin\frac{L_0\pi\varepsilon}{\sqrt{3}|\overrightarrow{R_b}|}$$
$$\left.+L_0\pi\varepsilon\cos\left(\frac{4\Delta y\pi}{\sqrt{3}|\overrightarrow{R_b}|}+\pi\left(\frac{2L_0\varepsilon}{\sqrt{3}|\overrightarrow{R_b}|}\right)\right)\sin\left(\pi\left(\frac{2L_0\varepsilon}{\sqrt{3}|\overrightarrow{R_b}|}\right)\right)\right], \quad \text{(S21)}$$

To analyze friction during sliding along the x-axis (with Δy constant), the friction force is calculated as,

$$f_{biaxial} = -\frac{\partial\Delta E_{biaxial}(\Delta x,\Delta y,\varepsilon,L_0)}{\partial\Delta x} = \frac{2\sqrt{3}|\overrightarrow{R_b}|(1+\varepsilon)^2}{\pi\varepsilon^2}\sin\frac{L_0\pi\varepsilon}{|\overrightarrow{R_b}|}\sin\frac{L_0\pi\varepsilon}{\sqrt{3}|\overrightarrow{R_b}|}\left(2\cos\left(\frac{\pi(2\Delta y+L_0\varepsilon)}{\sqrt{3}|\overrightarrow{R_b}|}\right)\sin\left(\frac{\pi(2\Delta x-L_0\varepsilon)}{|\overrightarrow{R_b}|}\right)\right). \quad \text{(S22)}$$

In addition to biaxial tension, uniaxial tension is also commonly observed in experiments, which significantly alters the moiré pattern. Consider a top-layer rectangular flake under uniaxial strain ε (>0) along the x-axis [Fig. S10(c)]. Since no deformation occurs in the y-direction, the density function is $\rho_0(x,y,\lambda)=\cos\left(\frac{2\pi x}{\lambda}\right)$ with $\lambda(\varepsilon)=|\overrightarrow{R_b}|\left(1+\frac{1}{\varepsilon}\right)$. When the flake translates by Δx in the x-direction, the moiré pattern shifts by $\Delta x\frac{\lambda}{|\overrightarrow{R_b}|(1+\varepsilon)}$ along the negative x-axis. Thus, the translated density function is $\rho(x,y,\Delta x,\Delta y,\lambda)=\rho_0\left(x+\Delta x\frac{\lambda}{|\overrightarrow{R_b}|(1+\varepsilon)},y,\lambda\right)$. Integrating this function over a flake region with length $L_0(1+\varepsilon)$ and height H_0 yields the translational interface energy function

$$\Delta E_{uniaxial} = \frac{H_0|\overrightarrow{R_b}|(1+\varepsilon)}{\pi\varepsilon}\cos\left[\pi\left(\frac{2\Delta x}{|\overrightarrow{R_b}|+|\overrightarrow{R_b}|\varepsilon}+\frac{L_0\varepsilon}{|\overrightarrow{R_b}|}\right)\right]\sin\left[\pi\frac{L_0\varepsilon}{|\overrightarrow{R_b}|}\right]. \quad \text{(S23)}$$

The frictional force during sliding along x-axis is

$$f_{uniaxial} = -\frac{\partial \Delta E_{uniaxial}(\Delta x, \varepsilon, L_0)}{\partial \Delta x} = \frac{2H_0}{\varepsilon} \sin\left[\pi\left(\frac{2\Delta x}{|\overrightarrow{R_b}| + |\overrightarrow{R_b}|\varepsilon} + \frac{L_0\varepsilon}{|\overrightarrow{R_b}|}\right)\right] \sin\left[\pi \frac{L_0\varepsilon}{|\overrightarrow{R_b}|}\right]. \quad \text{(S24)}$$

From Eqs. (S22) and (S24), the frictional force of the flake as a function of strain ε can be directly calculated for a specific initial flake size L_0.

We use this framework to interpret experimental results on CVD-grown graphene-graphene system [5]. Assuming $L_0 = 100nm$, we plot $F_{max} = max(f)$ for biaxial strain [Fig. S10(b) with $\Delta y = 0$] and uniaxial strain [Fig. S10(d)]. The results are consistent with experimental observations of strain-engineered superlubricity [5]. A sharp decrease in frictional force occurs when the strain reaches 0.2%, regardless of whether uniaxial or biaxial tension is applied. This matches the friction drop reported in Fig. 4 of Ref. [5]. Therefore, our theoretical method provides robust support for the experimental measurements of strain-engineered structural superlubricity.

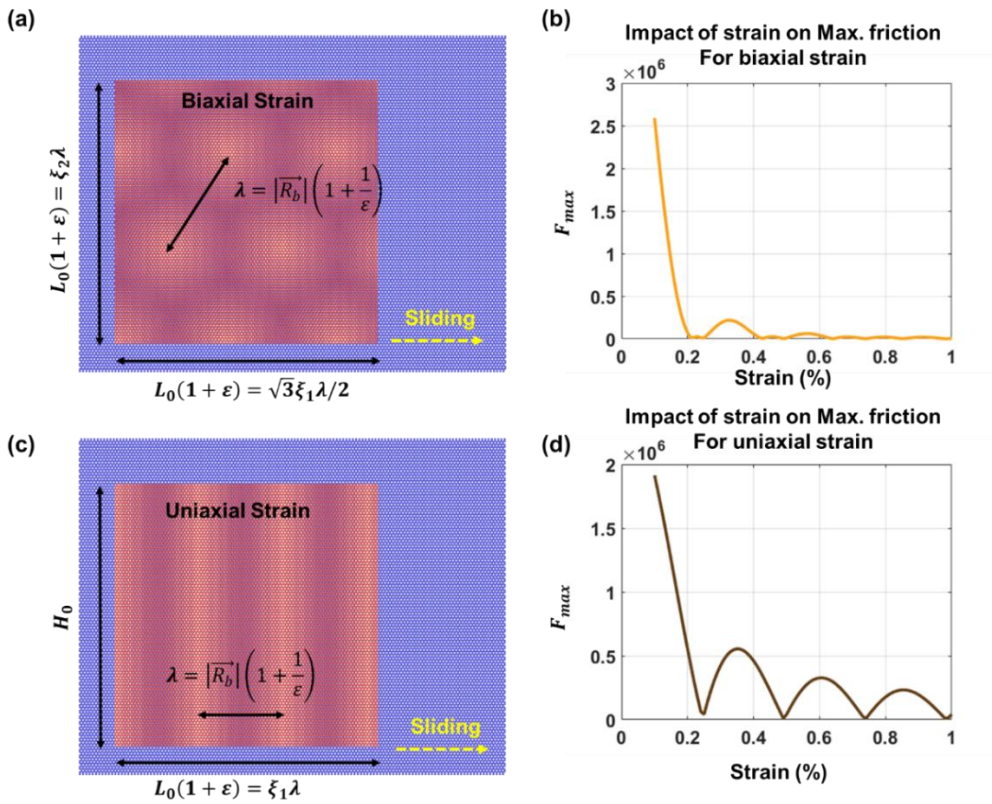

FIG. S10. (a) Schematic of biaxial strain application. (b) Maximal friction as a function of biaxial strain, calculated from Eq. (S22) with $\Delta y = 0$. (c) Schematic of uniaxial strain application. (d) Maximal friction as a function of uniaxial strain, calculated from Eq. (S24).

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